

CTMS-MAT-13: Numerical Methods**Problem Sheet 1. Released: 19 February 2026**

Exercise 1: Let $f(x) = \sin(\omega x)$ with some positive, real number ω .

- a) Write down the Taylor polynomials $p_n(x)$ of degree n for $f(x)$ around $c = 0$ for each of the following cases $n = 1, 2, 3, 4$.
- b) How large should n be so that $|\sin(x) - p_n(x)| < \varepsilon$ when $\varepsilon = 10^{-4}$ everywhere in the interval $[0, 1]$?

Exercise 2:

- a) Show that up to quadratic terms, the Taylor series for $\sqrt{1+x}$ about $c = 0$ can be written as:

$$p_2(x) = 1 + \frac{1}{2}x - \frac{1}{8}x^2.$$

- b) The remainder term is given by

$$\frac{1}{3!}(x-c)^3 f'''(\xi_x),$$

where ξ_x is between x and $c = 0$. Consider the maximum value of the remainder for all $x \in [0, 1]$, and, by simplifying, show that the upper bound to the remainder term is [1/16](#).

- c) Considering $x \in [0, \frac{1}{2}]$, what is the maximum value of the remainder term?

Exercise 3: Compute the Taylor series for $f(x) = e^{\cos(x)}$ around $c = 0$.

Exercise 4: Convert the following from one base to another and write down your calculations as an expansion:

- a) $(140)_{10}$ to $(\dots)_2$
- b) $(10.75)_{10}$ to $(\dots)_2$
- c) $(111.01001)_2$ to $(\dots)_8$
Hint: consider $(111)_2 + (010)_2 + (010)_2$ to get a three digit representation in base 10, then convert each digit from base 10 to base 8.

Exercise 4: Webcolors can be expressed with six base-16 (hexadecimal) digits (two each for the red, green and blue components, in that order) prefixed with #. The hexadecimal format uses sixteen distinct symbols, most often the symbols 0-9 to represent values 0 to 9, and A-F (or alternatively a-f) to represent values from ten to fifteen.

- a) How many separate shades are there in each channel of an RGB triplet and in total?
- b) How are black and white written in this format?
- c) Convert the hexadecimal colours [#008ce3](#), [#00204d](#) and [#db4f3d](#) into RGB triplets.
- d) CMYK colours encode four channels (cyan, magenta, yellow and black), taking values between 0-100 (inclusive). Are there more possible representations in the CMYK scheme than hexadecimal?

Exercise 5: The *ternary numerical system* uses three as its base. The number 100,000 in base 10 requires six digits. Its “radix economy” is therefore $10 \times 6 = 60$. For 10,000 the radix economy is 50.

- a) Show that, in base 2, the same number requires 14 digits, so its radix economy is $2 \times 14 = 28$.
- b) Show that, in base 3, it requires 9 digits, so its radix economy is $3 \times 9 = 27$.
- c) Show that, for 100,000 in base 2, the same number requires 17 digits, so its radix economy is $2 \times 17 = 34$.
- d) Again, for 100,000, show that, in base 3, the radix economy is 33.